

A Versatile Organozinc Approach to the Synthesis of Potential Organic Functional Building Blocks: 5-Substituted 3-Bromo-2-Methylthiophene Derivatives

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Photochromic compounds that undergo reversible transformation by appropriate stimulation have attracted much attention in the organic material fields, especially for photoswitching materials. A wide range of potential application for these photosensitive materials can be attributed to their stability, fatigue resistance, rapid response, and thermally irreversible properties.^{1,2} Of various functional photochromic compounds, diarylethenes are the most suitable molecules for the optoelectronic devices such as memories and switches.³

Numerous studies have been focused on the development of a high performance class of photosensitive molecules. One of the major breakthroughs was accomplished by introducing thiophene rings on both sides of ethene, which resulted in the formation of 1,2-dithienylethenes.⁴ Since then, this molecular structure has been considered as the most reliable basic platform to provide high performance in the devices operating based on photoresponsive functional materials.⁵ For the construction of the 1,2-bis(thien-3-yl)ethene-based photochromic molecules, two thiophene rings are generally connected by one of the following bridging units: maleic anhydride, maleimide, perfluorocyclopentene, cycloalkenes, or dihydrothiophene.^{1a,6}

To date, the most popular synthetic route for constructing the connection between the thiophene moiety and the ethene unit is the direct coupling reaction of lithiothiophene derivatives.⁷ More recently, Suzuki coupling has also been developed.⁸ To add more value into these methodologies, a larger variety of functionalized thiophene derivatives is strongly demanded. Among the thiophenes, 5-substituted 3-bromo-2-methylthiophenes have been the key intermediates for the assemble of the 1,2-bis(thien-3-yl)-system. To obtain the successful accomplishment of this synthetic journey, as described in the previous works mentioned above,⁷ the Suzuki coupling reaction conditions were predominately used. Interestingly, a recent work discovered another way to make the key intermediate, 5-arylated 3-bromo-2-methylthiophene, via the direct C–H activation of 3-bromo-2-methylthiophene.⁹

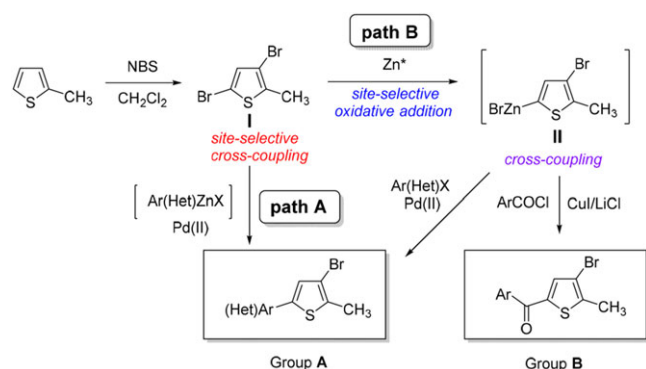
Taking the strategy above into account, exploring a more versatile synthetic approach toward 3-bromo-2-methylthiophene derivatives can provide ready access to a

wide range of flexible intermediates. Therefore, it is clearly in high demand. We herein would like to report a new route to synthesize 5-substituted 2-methylthiophenes.

The present study started with readily available 2-methylthiophene, which has an economical advantage over 3-bromo-2-methylthiophene, and the overall roadmap is depicted in Scheme 1. Bromination with *N*-bromosuccinimide was easily carried out using the procedure published in the literature, giving rise to the corresponding 3,5-dibrominated product. With the brominated thiophene, two different approaches were attempted for the preparation of 5-arylated thiophene derivatives: path **A** and path **B**, as shown in Scheme 1. First, the site-selective cross-coupling reactions of different organozinc reagents were examined (path **A**), and the results are summarized in Table 1. All of the organozinc reagents used in this study were prepared by the direct insertion of highly active zinc into the corresponding halocompounds.¹⁰

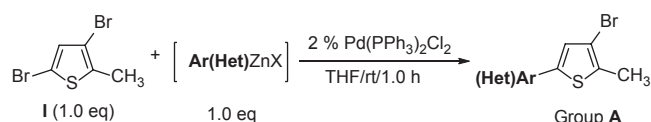
To our great delight, organozinc reagents were selectively coupled with 5-bromine on the thiophene ring in the presence of catalytic amounts of Pd(PPh₃)₂Cl₂ at room temperature. Organozincs from simple aryl and substituted aryl compounds were easily coupled to give the corresponding products (**1a–1c**) in moderate to good yields (entries 1–3, Table 1). More interestingly, heterocyclic arylzinc reagents were also effectively utilized for the selective couplings (entries 4–7, Table 1).

With the promising results obtained from the site-selective couplings of organozincs, more diversity of the strategy was then explored. This was accomplished utilizing 3-bromo-2-methylthien-5-ylzinc bromide (**II**). This essential organometallic molecule (**II**) was prepared by the regioselective direct insertion of highly active zinc into the carbon-bromine bond, as shown in Scheme 1 above. Treatment of 3,5-dibromo-2-methylthiophene with 1.2 equivalents of the active zinc prepared by the Rieke method resulted in the formation of **II**. Iodine quenching of the obtained thienylzinc bromide provided the corresponding 3-bromo-5-iodo-2-methylthiophene with an excellent yield (>95% by GC-MS). This observation provides evidence to rule out halogen dancing in the course of the organozinc preparation.



Scheme 1. Schematic diagram for 5-substituted 3-bromo-2-methylthiophenes

Table 1. Site-selective cross-coupling reaction of **I** with organozincs.



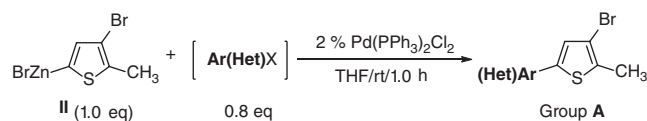
Entry	Ar(Het)ZnX	Product	Yield (%) ^a
1		1a	61
2		1b	68
3		1c	70
4		1d	75
5		1e	69
6		1f	78
7		1g	79

^a Isolated yield.

The applicability of **II** was first examined by the traditional Pd(II)-catalyzed cross-coupling reaction with various aryl and heteroaryl halocompounds. The results are summarized in Table 2. Once again, Pd(PPh₃)₂Cl₂ was very effective to complete the coupling at room temperature in THF. The corresponding 5-substituted 3-bromo-2-methylthiophene derivatives (**2a–2c**) were obtained in moderate to good isolated yields (entries 1–3, Table 2). Heterocyclic compounds were also good coupling partners to provide the corresponding 5-arylated thiophenes (**2d–2h**, entries 4–8, Table 2).

Utilization of the organozinc **II** was expanded to introduce a carbonyl group next to the thiophene ring. As depicted in

Table 2. Coupling reaction of **II** with arylhalides.



Entry	Ar(Het)X	Product	Yield (%) ^a
1		2a	73
2		2b	75
3		1b	84
4		1d	71
5		2c	70
6		2d	72
7		2e	78
8		2f	61

^a Isolated yield (based on halide).

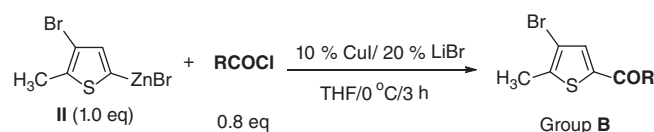
Scheme 1, group **B** compounds were successfully prepared via the coupling reaction of **II** with a wide range of acid chlorides. Not only aryl acid chlorides (entries 1–6), but also alkyl acid chlorides (entries 7–9) coupled with **II** under the normal coupling conditions (10 mol% CuI/20 mol% LiBr) in THF at 0 °C. Generally, the corresponding ketones (**3a–3i**, Table 3) were obtained in moderate to good yields.

As previously described in this work, various organozincs were selectively coupled with one of the bromines of 3,5-dibromo-2-methylthiophene. The selectivity of the new organozinc reagent **II** was then examined. It was treated with 2-bromo-4-iodopyridine under the same conditions used before, and the selectively coupled product (**4**) was obtained in 89% (Scheme 2).

A slight change of the ratio enables expansion of the scope of the expected utility of **II**. A dicoupling reaction of **II** was conducted with 2,6-dibromopyridine using similar reaction conditions, and the desired product (**5**) was obtained in 70% (Scheme 3).

For the preparation of a variety of thiophene derivatives, highly functionalized electrophiles are necessary as the coupling partners. To fulfill the high expectations, cross-coupling reactions of **II** with haloaromatic compounds containing relatively acidic protons were attempted. To this end, haloaromatic amines and phenols are reasonable candidates. By utilizing this strategy, 5-(4-aminophenyl)- and

Note

Table 3. Coupling of **II** with acid chlorides.

Entry	RCOCl	Product	Yield (%) ^a
1		3a	63
2		3b	74
3		3c	70
4		3d	78
5		3e	80
6		3f	62
7		3g	78
8		3h	68
9		3i	50

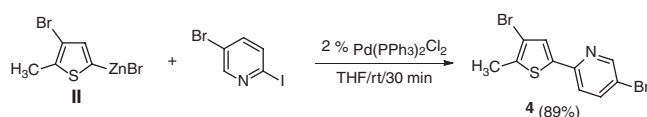
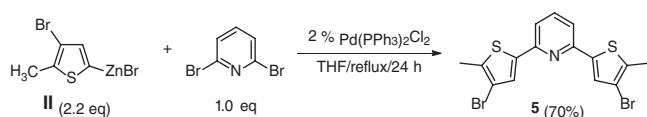
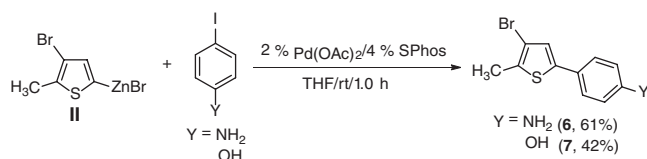
^a Isolated yield (based on RCOCl).

(4-hydroxyphenyl)-substituted thiophenes (**6** and **7**) were successfully prepared under mild conditions (Scheme 4).

In conclusion, a practical synthetic procedure for the preparation of 5-substituted 3-bromo-2-methylthiophene derivatives, which can potentially be utilized for the synthesis of photosensitive materials, has been demonstrated utilizing readily accessible 3,5-dibromo-2-methylthiophene (**I**). Both coupling reactions of appropriate organozinc reagents with **I**, and preparation of organozinc (**II**) from **I** and its application, turned out to be a superb approach for achieving the goal. Further examination of applications for the new organozinc reagent (**II**) is currently under way.

Experimental

Representative Procedures. Preparation of 3-bromo-2-methylthiophen-5-ylzinc bromide (II): In an oven-dried 250 mL round-bottomed flask equipped with a stir bar was added 4.70 g of active zinc (Zn*, 72.0 mmol). 3,5-Dibromo-2-methylthiophene (15.36 g, 60.0 mmol) was cannulated neat into the flask at room temperature. The resulting mixture was stirred for 1.0 h at room temperature. The whole mixture was

**Scheme 2.** Site-selective coupling of **II****Scheme 3.** Dicoupling of 2,6-dibromopyridine with **II****Scheme 4.** Coupling of **II** with aniline and phenol

settled down and then the supernatant was used for the subsequent coupling reactions.

Pd-catalyzed cross-coupling reaction: Preparation of 3-bromo-5-(4-cyanophenyl)-2-methylthiophene (**2a**): Into a 25 mL round-bottomed flask were added $\text{Pd}(\text{PPh}_3)_2\text{Cl}_2$ (0.035 g, 2.0 mol%), 4-bromobenzonitrile (0.36 g, 2.0 mmol), and 5.0 mL of 3-bromo-2-methylthiophen-5-ylzinc bromide (**II**) (0.5 M in THF, 2.5 mmol) under an argon atmosphere at room temperature. The resulting mixture was stirred at room temperature for 1.0 h. Quenched with 3 M HCl solution, then extracted with ethyl ether (10 mL $\times$ 3). Washed with saturated NaHCO_3 , $\text{Na}_2\text{S}_2\text{O}_3$ solution, and brine, and then dried over anhydrous MgSO_4 . Purification by column chromatography on silica gel (10% ethyl acetate/90% heptane) afforded 0.41 g of **2a** in 73% isolated yield as a yellow solid (m. p. 119–121 °C); ^1H NMR (CDCl_3 , 500 MHz) δ = 7.64 (d, J = 8.0 Hz, 2H), 7.58 (d, J = 8.0 Hz, 2H), 7.22 (s, 1H), 2.44 (s, 3H); ^{13}C NMR (CDCl_3 , 125 MHz) δ = 138.7, 137.6, 136.3, 132.8, 127.6, 125.4, 118.7, 110.8, 110.7, 15.0.

Cu-catalyzed cross-coupling reaction: Into a 25 mL round-bottomed flask were placed CuI (0.04 g, 10 mol%) and LiBr (0.04 g, 20 mol%). 3-Bromo-2-methylthiophen-5-ylzinc bromide (**II**) (5.0 mL, 0.5 M in THF, 2.5 mmol) was added into the flask under an argon atmosphere. Next, 4-bromobenzoyl chloride (0.44 g, 2.0 mmol) was slowly added via a syringe while being stirred at 0 °C. The resulting mixture was at 0 °C for 1 h. Quenched with 3.0 M HCl solution, then extracted with ethyl ether (10 mL $\times$ 3). Washed with saturated NaHCO_3 , $\text{Na}_2\text{S}_2\text{O}_3$ solution, and brine, and then dried over anhydrous MgSO_4 . Purification by column chromatography on silica gel (2% ethyl acetate/98% heptane) afforded 0.53 g of (4-bromo-5-methylthiophen-2-yl)-(4-bromophenyl) methanone (**3b**) in 74% isolated yield as a white solid (m. p. 108–109 °C); ^1H NMR (CDCl_3 , 500 MHz) δ = 7.70 (d, J = 7.5 Hz, 2H), 7.64 (d, J = 7.5 Hz, 2H), 7.42 (s, 1H), 2.49

(s, 3H); ^{13}C NMR (CDCl_3 , 125 MHz) δ = 185.7, 145.2, 139.4, 137.1, 136.0, 131.8, 130.5, 127.5, 110.6, 15.7.

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